

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Comparative Analysis

Code of Conduct:

I Cheredla Jasvina / 192324026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	

Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Annexure A

Comparative Analysis

1. CNNs vs. Vision Transformers (ViTs) for Object Detection Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

1. Compare CNNs and Vision Transformers in terms of:

- **Feature extraction capabilities for local vs. global context**
- **Performance with small or sparse objects**
- **Computational complexity and inference speed for embedded systems**

Answer :

Feature extraction: local vs. global context

CNNs are very effective at extracting local features such as edges, textures, shapes, and object boundaries. Their hierarchical structure allows them to gradually combine these local features to understand larger objects and scenes. This makes CNNs particularly useful for detecting objects such as traffic cones and animals.

Vision Transformers (ViTs) use self-attention to examine relationships between different parts of an image. This gives them a stronger ability to capture global context. For example, a ViT can relate a small object to the surrounding road, vehicles, and environment. However, this advantage usually comes with higher computational requirements.

Performance with small or sparse objects

CNNs generally perform well on small and sparse objects because their convolutional layers preserve local spatial information. Multi-scale feature extraction can further improve the detection of small obstacles.

ViTs can also detect small objects, but their performance can depend on the image resolution and patch size. If a small traffic cone or animal occupies only a few pixels, dividing the image

into large patches may cause important details to be lost.

Computational complexity and inference speed

CNNs are usually more computationally efficient and have fast inference speeds. Many lightweight CNN architectures are specifically designed for mobile and embedded devices, making them suitable for self-driving applications with limited computing power.

ViTs require more computation because self-attention needs to model relationships between image patches. Efficient ViT architectures have reduced this cost, but they can still require more memory and processing power than lightweight CNNs.

2. Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

Answer :

For this self-driving scenario, a lightweight CNN-based object detector would generally be the better choice. The system must detect rare obstacles in real time using embedded hardware and limited labeled data.

CNNs offer:

- Faster inference on resource-constrained hardware.
- Efficient extraction of local features.
- Good performance on small objects.
- Better data efficiency when training data is limited.
- Availability of lightweight architectures designed for real-time detection.

However, an efficient ViT or CNN–ViT hybrid could be considered if the vehicle has sufficient computational resources and global scene understanding is particularly important.

Effect of transfer learning

Transfer learning can significantly improve performance, especially when only a small number of labeled images are available. A model pretrained on a large image dataset can already learn general features such as edges, shapes, textures, and object patterns. The model can then be fine-tuned using images of traffic cones, animals, and other rare urban obstacles.

Transfer learning is particularly important for ViTs because they often benefit from large-scale pretraining.

Data augmentation can increase the diversity of training data and reduce overfitting. Useful techniques include:

- Random cropping and scaling
- Brightness and contrast changes
- Motion blur
- Rain, fog, and low-light simulation
- Random occlusion
- Horizontal flipping where appropriate
- Copy-paste augmentation of rare obstacles

For example, images of a traffic cone can be copied into different road scenes and under different lighting conditions. This helps the model learn to detect cones even when their appearance or surroundings change.

Conclusion

For this application, CNNs are generally better suited for real-time deployment on embedded systems, particularly when the dataset is small and the obstacles may be small or sparsely located. Transfer learning and data augmentation can further improve detection accuracy and robustness. Efficient ViTs are a promising alternative when more computational resources are available and global contextual information is important.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

1. Compare PCA-based shape priors and graph-based learning in terms of:

Ability to handle non-rigid deformations

o Robustness to varying mesh resolutions and noise

o Ease of implementation and computational cost

Answer :

Ability to handle non-rigid deformations

PCA-based shape priors learn the main patterns of variation from a set of aligned training shapes. The shape can then be represented using a small number of principal components. This works well for common facial deformations such as smiling, frowning, or opening the mouth. However, PCA mainly captures linear variations, so it may struggle with complex or highly non-rigid facial deformations.

Graph-based learning represents a 3D mesh as a graph, where vertices are nodes and mesh connections are edges. Graph neural networks can learn relationships between neighboring and distant facial regions. This allows them to model complex non-rigid deformations more effectively than a simple PCA model.

Robustness to varying mesh resolutions and noise

PCA-based methods generally assume that corresponding vertices have consistent meanings across meshes. Therefore, differences in topology and vertex density can make PCA difficult to apply directly. Meshes may need to be resampled or registered to a common topology first. PCA can also be affected by noise if noisy variations appear in the training data.

Graph-based methods are more flexible because they operate directly on the relationships within a mesh. They can handle different local structures and resolutions more naturally, although large differences in topology may still require preprocessing or specialized graph-matching techniques. Graph-based learning can also learn features that are more robust to noise.

Ease of implementation and computational cost

PCA is simple to implement and computationally inexpensive. Once the mean shape and principal components have been calculated, finding a low-dimensional representation is relatively fast. This makes PCA attractive for applications where simplicity and speed are important.

Graph-based learning is more complex to implement. It usually requires designing a graph neural network, preparing training data, and performing model training. It also requires greater computational resources, especially when processing high-resolution 3D meshes.

2. Analyze which approach would provide more accurate correspondences for realistic morphing and why.

Answer :

For the animation studio's problem, graph-based learning would generally provide more accurate correspondences, particularly because the meshes have different topology and vertex densities.

Realistic facial morphing requires accurate correspondence of important regions such as the eyes, nose, mouth, cheeks, and jaw. Graph-based learning can use geometric and neighborhood relationships to identify corresponding regions even when the meshes do not have identical vertex structures. It can also learn complex non-rigid facial **transformations**

from examples.

PCA-based shape priors would be easier and faster, but they work best when the meshes have already been brought into a consistent topology and vertex correspondence. If all faces have similar topology and the deformations are relatively regular, PCA can provide accurate results at a much lower computational cost.

Therefore, graph-based learning is the better choice for realistic morphing when mesh topology and resolution vary significantly. PCA can still be useful as a complementary technique—for example, as a low-dimensional shape prior or initialization for graph-based correspondence.

In summary: PCA is simpler, faster, and effective for standardized facial meshes, while graph-based learning is more flexible and generally more accurate for complex, non-rigid faces with varying mesh structures.